MANIPAL UNIVERSITY JAIPUR
(University under Section 2(f) of the UGC Act)

MANIPAL UNIVERSITY JAIPUR

Department of Computer and Communication Engineering

Course Hand-out

A. Basic Details:

Programme Name:	Computer and Communication Engineering
Course Name:	Digital Design and Computer Architecture
Course Code:	CCE 2101
LTPC *(Lecture Tutorial Practical Credits):*	3 0 1 4
Session:	Jul-Nov 2025
Class:	II year
Course Coordinator:	Dr. Kusum Lata Jain
Course Instructor(s):	Dr. Kusum Lata Jain, Dr. Arvind Dhaka
Additional Practitioner(s) – if any *(Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.):*	-

B. Introduction: This course is offered by Dept. of Computer and Communication Engineering for third semester students. The core objective of this course is to describe the general organization and architecture of a computer system. It covers in detail various functional units of a computer system, machine instructions, addressing techniques and instruction sequencing. It provides a detailed coverage of logic circuits to perform various arithmetic operations and use of pipelining in the design of high-performance processors.

C. Course Outcomes:

CO Statement	*CO*	*Bloom's Level*	*Target Attainment %*	*Target Attainment level*
1. Design Sequential and Combination Circuits using digital components for computer system.	2101.1	L3	75	2
2. Describe various data representation and formulate assembly language	2101.2	L2	65	1

programs for a given high level language construct.				
3. Analyse the design of fast arithmetic circuits.	2101.3	L4	75	2
4. Evaluate the performance of CPU, memory, and I/O operations.	2101.4	L4	75	2
5. Build the required skills to read and research the current literature in computer architecture.	2101.5	L3	75	2

D. Program Outcomes and Program Specific Outcomes:

[PO.1] Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems

[PO.2] Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

[PO.3] Design/development of solutions: Design solutions for complex engineering problems and design system components or processes_that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

[PO.4] Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

[PO.5] Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools_including prediction and modeling to complex engineering activities with an understanding of the limitations.

[PO.6] The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues_and the consequent responsibilities relevant to the professional engineering practice.

[PO.7] Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

[PO.8] Ethics: Apply ethical principles and commit to professional ethics_and responsibilities and norms of the engineering practices

[PO.9] Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

[PO.10] Communication: Communicate effectively_on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions

[PO.11] Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

[PO.12] Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

At the end of the B Tech CCE program, the student:

Program Specific Outcome-PSOs

[PSO.1] Imbibe the basic concepts and applications of computer-based Communication or networking, information sharing, signal processing, web-based systems, smart devices, and communication technology.

[PSO.2] Investigate prominent areas in the field of Computer and Communication Engineering to provide feasible solutions.

[PSO.3] Apply the contextual knowledge in the field of Computing and Communication to assess social, health, safety, and security issues relevant to professional engineering

E. Assessment Plan: Clearly write the criteria, its description, and associated marks for assessment of student achievements. In addition, the attendance requirements, assignments need to be mentioned in this section.

Criteria	Description	Maximum Marks
Internal Assessment (Formative and Summative)	Mid-Term Examination (Close)	30
	Class Work Sessional (CWS): 5 Quiz and 2 assignments	30
End Term Exam (Summative)	End Term Exam (Close)	40
	Total	100

F. Syllabus: Basic Structure of Computers: Computer Types, Functional Units, Basic Operational Concepts, Software, Performance; Digital logic circuits: logic gates, Boolean algebra, map simplification, combinational circuits, flip-flops, sequential circuits; Digital Components: Integrated circuits, decoders, multiplexers, registers, shift registers, binary counters; Machine instructions and programs: numbers, arithmetic operations and characters, memory locations and addresses, instructions and instruction sequencing, addressing modes, assembly language, additional instructions, encoding of machine instructions; micro-operation and their RTL specification Arithmetic: addition and subtraction of signed numbers, design of fast adders, multiplication of positive numbers, signed operand multiplication, fast multiplication, integer division, floating point numbers and operations; Introduction to input/output processing: programmed controlled i/o transfer, interrupt controlled I/O transfer, DMA controller; Pipelining and pipeline hazards: design issues of pipeline architecture; Instruction level parallelism and advanced issues.

References:

1. M. M. Mano, Computer System Architecture, (3e), Pearson Education, 2014.

2. C. Hamacher, Z. Vranesic, S. Zaky, Computer Organization, (6e), McGraw Hill, 2011.

3. J. P. Hayes, Computer Architecture and Organization, (3e), McGraw Hill, 2017.

4. T. L. Floyd, Digital Fundamentals, (10e), Pearson Education, 2014.

5. W. Stallings, Computer Organization and Architecture–Designing for Performance, (8e), Pearson Education, 2010.

G. Lecture Plan:

Lecture	Topics	Session Outcome	Corresponding CO	Mode of Delivery	Mode Of Assessing CO
1	Discussion of Course handout, course Outcome	Lecture plan	-	Lecture based teaching-learning	NA
2	Computer Types, Functional Units	Understand basic computer system	**2101.1**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
3	Basic Operational Concepts	Understand basic operational Concept	**2101.1**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
4	Software, Performance	Explain software and calculate performance of a system	**2101.1**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
5	Logic gates	Understand logic gates	**2101.1**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
6	Boolean algebra,	Explain Boolean algebra	**2101.1**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
7	map simplification,	Perform Map simplification using K-Map	**2101.1**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
8	combinational circuits	Design Combination Circuits (Adder)	**2101.1**	Learning through problem-solving	Mid Term, Assignment, Quiz & End Term
9	Design combinational circuits	Design Combination Circuits (others)		Learning through problem-solving	Mid Term, Assignment, Quiz & End Term
10	Basic of Flip-Flops	Understand Flip-Flops	**2101.1**	Learning through problem-solving	

11	Flip-Flop	Explain type of flipflops	**2101.1**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
12	sequential circuits	Describe Sequential Circuits.	**2101.1**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
13	Integrated circuits, decoders	Understand integrated circuits, and decoder	**2101.1**	Learning through problem-solving	Mid Term, Assignment, Quiz & End Term
14	Multiplexers	Explain multiplexer		Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
15	Registers, shift registers, binary counters	Explain type of register and boundary counters	**2101.1**	Lecture based teaching-learning	
16	Numbers, binary Arithmetic Operations and Characters	Represent numbers and perform binary arithmetic operation on numbers	**2101.1**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
17	Memory Locations and Addresses,	Understand memory location and operations	**2101.2**	Flipped Classroom	Mid Term, Assignment, Quiz & End Term
18	Memory Operations	Understand Memory Operations	**2101.2**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
19	Instructions and Instruction Sequencing	Describe instructions and instruction sequence	**2101.2**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
20	addressing modes	Explain different addressing modes	**2101.2 2101.3**	Lecture based teaching-learning	Mid Term, Assignment, Quiz & End Term
21	assembly language, additional instructions, encoding of machine instructions	Convert program into assembly language	**2101.2**	Learning through problem-solving	Mid Term, Assignment, Quiz & End Term

22	Addition and Subtraction of Signed Numbers	Perform addition and multiplication on signed numbers	**2101.2**	Learning through problem-solving	Mid Term, Assignment, Quiz & End Term
23	Design of Fast Adders	Design and analysis of Fast Adder			Mid Term, Assignment, Quiz & End Term
24	Carry Look Ahead Adders	Analysis of carry look ahead adder	**2101.3**	Learning through problem-solving	Mid Term, Assignment, Quiz & End Term
25	Multiplication of Positive Numbers-Array	Perform multiplication on numbers		Learning through problem-solving	Mid Term, Assignment, Quiz & End Term
26	Sequential Circuit	Perform booth algorithm for singed numbers	**2101.3**	Learning through problem-solving	Mid Term, Assignment, Quiz & End Term
27	Signed Operand Multiplication-Booth Algorithm	Demonstrate fast multiplication by bit pair recording-I	**2101.3**	Learning through problem-solving	Mid Term, Assignment, Quiz & End Term
	MID TREM EXAMINATION				
28	Fast Multiplication-Bit Pair Recoding of Multipliers	Demonstrate fast multiplication by bit pair recording-II	**2101.3**	Learning through problem-solving	Assignment, Quiz & End Term
29	Fast Multiplication-Bit Pair Recoding of Multipliers	Calculate sum by carry save addition		Learning through problem-solving	Assignment, Quiz & End Term
30	Carry-save addition of summands	Demonstrate Integer Division	**2101.3**	Learning through problem-solving	Assignment, Quiz & End Term
31	Remedial Class 1	Revision 1	**2101.3** 2101.5	Group- teaching and learning	Assignment, Quiz & End Term
32	Remedial Class 2	Revision 2	**2101.3**	Group- teaching and learning	Assignment, Quiz & End Term
33	Integer Division	Restoring division for unsigned numbers	**2101.3**	Learning through problem-solving	Assignment, Quiz & End Term
34	Integer Division- restoring	Perform non-restoring method for singed numbers	**2101.3**	Learning through problem-solving	Assignment, Quiz & End Term

35	Integer Division- Nonrestoring	Discuss standard to represent floating point numbers	**2101.3**	Technology based learning	Assignment, Quiz & End Term
36	Floating Point Numbers & Operation- Standards	Perform arithmetic operation on floating point numbers	**2101.4 2101.5**	Learning through problem-solving	Assignment, Quiz & End Term
37	Floating Point Numbers & Operation- Standards	Perform arithmetic operation on floating point numbers	**2101.4 2101.5**	Learning through problem-solving	Assignment, Quiz & End Term
38	Arithmetic Operations on Floating Point Numbers	Perform and demonstrate arithmetic operation on floating point numbers	**2101.4 2101.5**	Learning through problem-solving	Assignment, Quiz & End Term
39	Introduction to Input/Output Processing	Discuss Input/Ouput Processing	**2101.4**	Technology based learning	Assignment, Quiz & End Term
40	instruction interpretation and execution	Explain instruction interpretation and execution	**2101.4**	Individual learning/ self-study	Assignment, Quiz & End Term
41	micro-operation and their RTL specification	Discuss Micro Operation	**2101.4**	Lecture based teaching-learning	Assignment, Quiz & End Term
42	programmed controlled i/o transfer,	Explain Programmed Controlled transfer	**2101.4**	Lecture based teaching-learning	Assignment, Quiz & End Term
43	interrupt controlled I/O transfer,	Explain interrupt Controlled transfer	**2101.4**	Lecture based teaching-learning	Assignment, Quiz & End Term
44	Accessing I/O Devices,	Explain access methods for I/O devices	**2101.4**	Lecture based teaching-learning	Assignment, Quiz & End Term
45	Handling Multiple Devices, Controlling Device Requests, Exceptions	Discuss controlling device and exceptions	**2101.4**	Lecture based teaching-learning	Assignment, Quiz & End Term
46	Direct Memory Access,	Understating DMA	**2101.4**	Lecture based teaching-learning	Assignment, Quiz & End Term
47	DMA controller;	Discuss DMA Controller	**2101.4**	Individual learning/ self-study	Assignment, Quiz & End Term
48	Design issues of pipeline architecture.	Discuss pipeline	**2101.4**	Technology based learning	Assignment, Quiz & End Term
49	Pipelining	Explain data Hazards	**2101.4**	Technology based learning	Assignment, Quiz & End Term

50	Data Hazards	Demonstrate Instruction	**2101.4**	Lecture based teaching-learning	Assignment, Quiz & End Term
51	Instruction Scheduling: Static and Dynamic	Discuss instruction Scheduling	**2101.4**	Lecture based teaching-learning	Assignment, Quiz & End Term
52	Instruction level parallelism	Exaplin instruction level parallelism	**2101.4**	Lecture based teaching-learning	Assignment, Quiz & End Term
END TERM EXAMINATION					

H. Learner centric activities:

Outside class engagement activities (To complete the 30 Notional hours for each credit as per NCrF guidelines):

Type of Activity	Difficulty Level	Hours Engaged
Step-by-step problem-solving exercises	Low	5
Higher-order thinking assignments	Low	5
online study materials on the subjects from top experts in the domain	HIGH	5

I. Outside Class Engagement Activities:

- Participating in academic clubs, technical competitions, or seminars
- Collaborating with faculty or peers on research ideas or innovative prototypes
- Attending faculty-guided doubt-clearing or reinforcement sessions
- Watching recommended video lectures, tutorials, or educational webinars
- Participating in group study circles or peer-learning sessions
- Writing short reflective summaries or maintaining learning journals
- Contributing to departmental blogs, newsletters, or student magazines

J. Course Articulation Matrix: (Mapping of COs with POs)

CO	STATEMENT	CORRELATION WITH PROGRAM OUTCOMES												CORRELATION WITH PROGRAM SPECIFIC OUTCOMES		
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
2101.1	Design Sequential and Combination Circuits using digital components for computer system.	2	1										1	2	1	
2101.2	Describe various data representation and formulate assembly language programs for a given high level language construct	3	2	1									1	1	1	
2101.3	Analyse the design of fast arithmetic circuits.	2	2	1									1	1		
2101.4	Evaluate the performance of CPU, memory, and I/O operations	3	2										1		1	1
2101.5	Build the required skills to read and research the current literature in computer architecture.	3	2	1									2	1	1	2

1- Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation